

Software Project Management: Complete Study Guide

MSc Computer Science - CSE 531

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Table of Contents

- Chapter 1: Feasibility Study in Software Engineering
 - Chapter 2: Managing Goals, Time, and Cost
 - Chapter 3: Risk and Quality Management
-

Chapter 1: Feasibility Study in Software Engineering

1.1 Introduction to Feasibility Study

A **feasibility study** is the initial step in the software development life cycle (SDLC) that evaluates whether a proposed software project is practical, achievable, and worth investing in. The primary objectives include:

- Assessing the viability of a software project
- Identifying potential risks and constraints
- Supporting decision-making on whether to proceed, modify, or cancel the project

1.1.1 Outcome: Feasibility Report

A feasibility report summarizes findings and provides a clear go/no-go decision. It includes:

- Project overview and objectives
- Findings for each feasibility type
- Cost-benefit and break-even analysis
- Risk identification and mitigation strategies
- Final recommendation

1.2 Types of Feasibility Analysis

Software engineering feasibility is analyzed across five main dimensions:

1.2.1 Technical Feasibility

Definition: Evaluates whether the technology, tools, and skills required for the project are available and capable of developing the proposed system.

Key Focus Areas:

- Availability of hardware and software resources
- Compatibility with existing systems
- Technical skills of the development team
- System performance requirements (speed, storage, reliability, scalability)
- Technology risk (e.g., new, untested tools)

Example: A company wants to implement an AI-based defect prediction tool. If the team has experience with Python and ML frameworks, and the infrastructure can handle large datasets, the project is technically feasible.

1.2.2 Economic Feasibility

Definition: Determines whether the project is financially worthwhile by comparing expected costs and benefits to ensure a good return on investment (ROI).

Key Focus Areas:

- Development costs (salaries, equipment, software licenses)
- Operational costs (maintenance, hosting, updates)
- Tangible benefits (revenue, reduced labor cost, faster operations)
- Intangible benefits (user satisfaction, reputation, data accuracy)
- Break-even point (BEP) when benefits equal costs

1.2.3 Operational Feasibility

Definition: Checks whether the system will work in the real-world environment and be accepted by end users.

Key Focus Areas:

- System fit into daily business operations
- User attitude, motivation, and adaptability
- Organizational changes required
- Training and support requirements
- Ease of use and reliability

Example: If employees resist moving from manual to automated systems due to fear of job loss, the system may not be operationally feasible, even if technically possible.

1.2.4 Legal Feasibility

Definition: Ensures the project complies with all legal and regulatory requirements.

Key Focus Areas:

- Software licensing and copyright issues
- Data privacy and protection laws (e.g., GDPR, HIPAA)
- Intellectual property rights
- Contractual obligations with clients or vendors
- Industry standards and certifications

Example: Hospital software storing patient information must comply with data protection laws; otherwise, it fails legal feasibility.

1.2.5 Schedule Feasibility

Definition: Determines whether the project can be completed within the given timeframe using available resources.

Key Focus Areas:

- Realistic project timeline
- Task dependencies and milestones
- Availability of skilled staff
- Time for testing and deployment
- Impact of delays on the organization

Example: If a system must be deployed before a government audit in 4 months but needs 6 months of development, it fails schedule feasibility.

1.3 Economic Feasibility: Mathematical Methods

1.3.1 Cost-Benefit Ratio (CBR)

Definition: CBR compares the total expected benefits of a project to its total expected costs, indicating how much benefit you get per unit of cost.

Formula:

$$\text{CBR} = \frac{\text{Total Benefits}}{\text{Total Costs}}$$

Decision Criteria:

- If $\text{CBR} > 1$: Project is profitable (feasible)
- If $\text{CBR} = 1$: Project breaks even
- If $\text{CBR} < 1$: Project is not financially viable

Example 1.1: CBR Calculation

A software project has the following financial data:

- Total Cost: \$100,000
- Total Benefit: \$150,000

Solution:

$$\text{CBR} = \frac{150,000}{100,000} = 1.5$$

Interpretation: For every dollar invested, the project returns \$1.50. Since $\text{CBR} > 1$, the project is economically feasible.

1.3.2 Break-Even Point (BEP)

Definition: Break-even analysis identifies the point at which total cost equals total revenue or benefits (no profit, no loss).

Formula for Break-Even Point (Units):

$$\text{BEP} = \frac{\text{Fixed Costs}}{\text{Selling Price per Unit} - \text{Variable Cost per Unit}}$$

Example 1.2: BEP Calculation

A software company develops a product with:

- Fixed Cost: \$50,000
- Selling Price per License: \$10
- Variable Cost per License: \$5

Solution:

$$\text{BEP} = \frac{50,000}{10 - 5} = \frac{50,000}{5} = 10,000 \text{ licenses}$$

Interpretation: The company needs to sell 10,000 licenses to cover all costs. Beyond this point, the project generates profit.

Example 1.3: BEP with Annual Benefits

- Development Cost: \$100,000
- Expected Annual Savings: \$40,000

Solution:

$$\text{BEP (years)} = \frac{100,000}{40,000} = 2.5 \text{ years}$$

Interpretation: The project will break even in 2.5 years, making it economically feasible.

1.3.3 Net Present Value (NPV)

Definition: NPV measures the current (present) value of all future cash inflows and outflows of a project. It considers the **time value of money**, meaning money today is worth more than the same amount in the future due to inflation and opportunity cost.

Formula:

$$\text{NPV} = \sum_{t=0}^n \frac{B_t - C_t}{(1 + r)^t}$$

Where:

- B_t = Benefits (cash inflow) in year t
- C_t = Costs (cash outflow) in year t
- r = Discount rate (interest or inflation rate)
- n = Number of years
- t = Time period (year)

Decision Criteria:

- If $\text{NPV} > 0$: Project is profitable
- If $\text{NPV} = 0$: Project breaks even
- If $\text{NPV} < 0$: Project loses money

Example 1.4: NPV Calculation

A project has the following data:

- Initial Investment: \$100,000 (Year 0)
- Expected Benefits: \$40,000 per year for 3 years
- Discount Rate: 10% ($r = 0.10$)

Solution:

Year 0:

$$PV_0 = -100,000$$

Year 1:

$$PV_1 = \frac{40,000}{(1 + 0.10)^1} = \frac{40,000}{1.10} = 36,364$$

Year 2:

$$PV_2 = \frac{40,000}{(1 + 0.10)^2} = \frac{40,000}{1.21} = 33,058$$

Year 3:

$$PV_3 = \frac{40,000}{(1 + 0.10)^3} = \frac{40,000}{1.331} = 30,053$$

Total NPV:

$$NPV = -100,000 + 36,364 + 33,058 + 30,053 = -525$$

Interpretation: NPV is slightly negative (-\$525), indicating the project marginally loses money when considering the time value of money. The project manager should reconsider or negotiate better terms.

Example 1.5: NPV with Positive Outcome

Modified scenario:

- Initial Investment: \$100,000
- Expected Benefits: \$50,000 per year for 3 years
- Discount Rate: 10%

Solution:

$$PV_1 = \frac{50,000}{1.10} = 45,455$$

$$PV_2 = \frac{50,000}{1.21} = 41,322$$

$$PV_3 = \frac{50,000}{1.331} = 37,566$$

$$NPV = -100,000 + 45,455 + 41,322 + 37,566 = 24,343$$

Interpretation: NPV is positive (\$24,343), indicating the project is profitable and economically feasible.

Chapter 2: Managing Goals, Time, and Cost

2.1 Project Value Assessment

2.1.1 Financial Methods

Three primary financial methods assess project value:

1. **Payback Period:** Measures how long it takes to return the project's investment
2. **Return of Investment (ROI):** Measures efficiency of investment
3. **Net Present Value (NPV):** Considers time value of money

2.1.2 Return of Investment (ROI)

Definition: ROI measures how much return we get for each dollar invested.

Formula:

$$\text{Annual Profit} = \frac{\text{Total Profit}}{\text{Project Duration (years)}}$$

$$\text{ROI} = \frac{\text{Annual Profit}}{\text{Total Project Expenses}} \times 100\%$$

Example 2.1: ROI Calculation

Project A:

- Total Expenses: \$100,000

- Total Profit over 5 years: \$50,000

Solution:

$$\text{Annual Profit} = \frac{50,000}{5} = 10,000$$

$$\text{ROI} = \frac{10,000}{100,000} \times 100\% = 10\%$$

Note: The original material states ROI of 11% for Project A. This may include additional considerations not shown in simplified calculation.

Project B:

- Total Expenses: \$150,000
- Total Profit over 5 years: \$285,000

Solution:

$$\text{Annual Profit} = \frac{285,000}{5} = 57,000$$

$$\text{ROI} = \frac{57,000}{150,000} \times 100\% = 38\%$$

Interpretation: Project B has a higher ROI (38%) compared to Project A (approximately 10-11%), making it more financially attractive despite higher initial costs.

2.2 Score Matrix Method

Definition: A score matrix evaluates projects based on multiple criteria beyond financial metrics, capturing qualitative and strategic factors.

2.2.1 Score Matrix Construction

1. List project criteria (both desirable and undesirable aspects)
2. Assign weights to each criterion
 - Desirable features: Positive weights (e.g., +1 to +5)
 - Undesirable features: Negative weights (e.g., -1 to -5)

3. Score each project on each criterion (e.g., 1 to 5 scale)
4. Multiply scores by weights
5. Sum all weighted scores

Formula:

$$\text{Total Score} = \sum_{i=1}^n (\text{Criterion Score}_i \times \text{Criterion Weight}_i)$$

Example 2.2: Score Matrix Comparison

Criterion	Weight	Project 1		Project 2	
		Score	Weighted	Score	Weighted
Technical Feasibility	+5	5	25	4	20
Cost Efficiency	+4	3	12	5	20
Market Demand	+5	5	25	3	15
Implementation Time	+3	4	12	4	12
Risk Level	-4	2	-8	4	-16
Total			66		51

Interpretation: Project 1 scores higher (66 vs. 51), making it the preferred choice despite not being superior in all individual categories.

2.3 SWOT Analysis

Definition: SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis evaluates internal and external factors affecting project success.

2.3.1 SWOT Components

- **Strengths (Internal, Positive):** Project advantages and capabilities
- **Weaknesses (Internal, Negative):** Project limitations and vulnerabilities

- **Opportunities (External, Positive):** Favorable external conditions
- **Threats (External, Negative):** Unfavorable external conditions

Example 2.3: SWOT Analysis for Enrollment Software

Strengths	Weaknesses
- User-friendly interface - Automation of tasks - Strong data security - Scalability - Customizable features	- High initial cost - Complex setup - Limited system integration - Dependence on internet connectivity
Opportunities	Threats
- Growing demand for online education - Partnerships with institutions - Market expansion - Tech advancements (AI)	- Strong competition - Data privacy regulations - Technology obsolescence - Cybersecurity risks

2.4 Effort Estimation Techniques

2.4.1 Three-Point Estimation (PERT)

Definition: PERT (Program Evaluation and Review Technique) uses three estimates to account for uncertainty in task duration.

Formula:

$$\text{Expected Time (TE)} = \frac{t_O + 4t_M + t_P}{6}$$

Where:

- t_O = Optimistic time (best-case scenario)
- t_M = Most likely time (realistic estimate)

- t_P = Pessimistic time (worst-case scenario)

Example 2.4: PERT Calculation

Activity A:

- Optimistic Time: 5 days
- Most Likely Time: 6 days
- Pessimistic Time: 7 days

Solution:

$$TE = \frac{5 + 4(6) + 7}{6} = \frac{5 + 24 + 7}{6} = \frac{36}{6} = 6 \text{ days}$$

Example 2.5: Multiple Activities

Activity	t_O	t_M	t_P	TE
A	5	6	7	6
B	2	3	4	3
C	3	4	5	4
D	1	2	3	2
E	2	3	4	3
F	4	5	6	5
G	2	3	4	3
H	4	5	6	5
I	4	5	6	5
J	2	3	4	3

2.5 Function Points (FP)

Definition: Function Points measure system size based on functionality rather than lines of code.

2.5.1 Functional Characteristics

Five functional measures:

1. **User Inputs:** Elements requiring user input
2. **User Outputs:** Elements producing output
3. **User Inquiries:** Inputs generating software response
4. **Internal Logical Files:** Files created and used dynamically
5. **External Interfaces:** Files connecting to external systems

2.5.2 Complexity Classification

Each characteristic is classified as:

- Simple (lower weight)
- Average (medium weight)
- Complex (higher weight)

2.5.3 Weight Matrix

Component	Simple	Average	Complex
User Inputs	3	4	6
User Outputs	4	5	7
User Inquiries	3	4	6
Internal Files	7	10	15
External Interfaces	5	7	10

2.5.4 Unadjusted Function Points (UFP)

Formula:

$$\text{UFP} = \sum (\text{Count}_i \times \text{Weight}_i)$$

Example 2.6: UFP Calculation

A system has:

- 10 simple user inputs
- 5 average user outputs
- 3 simple user inquiries

- 2 average internal files
- 1 complex external interface

Solution:

$$\text{UFP} = (10 \times 3) + (5 \times 5) + (3 \times 3) + (2 \times 10) + (1 \times 10)$$

$$\text{UFP} = 30 + 25 + 9 + 20 + 10 = 94$$

2.5.5 Value Adjustment Factor (VAF)

Formula:

$$\text{VAF} = \frac{65 + \sum_{i=1}^{14} \text{Influence}_i}{100}$$

Where influence scores range from 0 (irrelevant) to 5 (very influential) for 14 nonfunctional characteristics.

Example 2.7: VAF Calculation

Sum of 14 nonfunctional influences = 35

$$\text{VAF} = \frac{65 + 35}{100} = \frac{100}{100} = 1.00$$

2.5.6 Adjusted Function Points (AFP)

Formula:

$$\text{AFP} = \text{UFP} \times \text{VAF}$$

Using Example 2.6 and 2.7:

$$\text{AFP} = 94 \times 1.00 = 94$$

2.6 COCOMO Model

Definition: COCOMO (Constructive Cost Model) estimates effort, time, and cost based on project size measured in thousands of lines of code (KSLOC).

2.6.1 COCOMO I (COCOMO 81) - Basic Model

Project Types

1. **Organic:** Small, simple projects with experienced teams
2. **Semi-Detached:** Intermediate complexity with mixed experience
3. **Embedded:** Complex projects with stringent requirements

Basic COCOMO Formulas

Effort (PM - Person-Months):

$$PM = A_{PM} \times (KSLOC)^{B_{PM}}$$

Development Time (TDEV - Months):

$$TDEV = A_{TDEV} \times (PM)^{B_{TDEV}}$$

Team Size:

$$TEAM = \frac{PM}{TDEV}$$

COCOMO Coefficients

Project Type	A_{PM}	B_{PM}	A_{TDEV}	B_{TDEV}
Organic	2.4	1.05	2.5	0.38
Semi-Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Example 2.8: COCOMO Calculation (Semi-Detached)

Given:

- Project Size: 10 KSLOC
- Project Type: Semi-Detached

Solution:

Step 1: Calculate Effort (PM)

$$PM = 3.0 \times (10)^{1.12}$$

$$10^{1.12} = 10^1 \times 10^{0.12} = 10 \times 1.318 = 13.18$$

$$PM = 3.0 \times 13.18 = 39.54 \text{ person-months}$$

Step 2: Calculate Development Time (TDEV)

$$TDEV = 2.5 \times (39.54)^{0.35}$$

$$39.54^{0.35} = 3.408$$

$$TDEV = 2.5 \times 3.408 = 8.52 \text{ months}$$

Step 3: Calculate Team Size

$$TEAM = \frac{39.54}{8.52} = 4.64 \approx 5 \text{ people}$$

Interpretation: The project requires approximately 40 person-months of effort, will take about 8.5 months to complete, and needs a team of 5 people.

Example 2.9: COCOMO Calculation (Organic)

Given:

- Project Size: 5 KSLOC
- Project Type: Organic

Solution:

Effort:

$$PM = 2.4 \times (5)^{1.05} = 2.4 \times 5.18 = 12.43 \text{ person-months}$$

Development Time:

$$TDEV = 2.5 \times (12.43)^{0.38} = 2.5 \times 2.60 = 6.5 \text{ months}$$

Team Size:

$$\text{TEAM} = \frac{12.43}{6.5} = 1.91 \approx 2 \text{ people}$$

Example 2.10: COCOMO Calculation (Embedded)

Given:

- Project Size: 20 KSLOC
- Project Type: Embedded

Solution:

Effort:

$$\text{PM} = 3.6 \times (20)^{1.20} = 3.6 \times 30.56 = 110.02 \text{ person-months}$$

Development Time:

$$\text{TDEV} = 2.5 \times (110.02)^{0.32} = 2.5 \times 4.26 = 10.65 \text{ months}$$

Team Size:

$$\text{TEAM} = \frac{110.02}{10.65} = 10.33 \approx 10 \text{ people}$$

2.6.2 COCOMO II

COCOMO II addresses modern software development approaches including object-oriented, agile, and component-based development.

COCOMO II Models

1. **Application Composition Model:** For prototyping with reusable components
 2. **Early Design Model:** During system design phase using function points
 3. **Reuse Model:** For integrating reusable components
 4. **Post-Architecture Model:** Most detailed estimates after architecture is defined
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2.7 Critical Path Method (CPM)

Definition: The Critical Path is the longest sequence of dependent activities that determines the minimum project duration. Any delay on the critical path delays the entire project.

2.7.1 Key Concepts

- **Early Start (ES):** Earliest time an activity can begin
- **Early Finish (EF):** Earliest time an activity can complete
- **Late Start (LS):** Latest time an activity can begin without delaying project
- **Late Finish (LF):** Latest time an activity can finish without delaying project
- **Slack (Float):** Amount of time an activity can be delayed without affecting project completion

2.7.2 Critical Path Formulas

Forward Calculation (ES and EF):

$$EF = ES + \text{Duration}$$

For activities with multiple predecessors:

$$ES = \max(\text{EF of all predecessors})$$

Backward Calculation (LS and LF):

$$LS = LF - \text{Duration}$$

For activities with multiple successors:

$$LF = \min(\text{LS of all successors})$$

Slack Calculation:

$$\text{Slack} = LS - ES = LF - EF$$

Critical Path: All activities where Slack = 0

2.7.3 Critical Path Example

Example 2.11: Complete Critical Path Analysis

Activity Network:

Activity	Predecessors	Duration (days)
A	-	6
B	-	3
C	-	4
D	A	2
E	B	3
F	C	5
G	C	3
H	E, F	5
I	D	5
J	G, H	3

Step 1: Forward Pass (Calculate ES and EF)

Activity	ES	Duration	EF
A	0	6	6
B	0	3	3
C	0	4	4
D	6	2	8
E	3	3	6
F	4	5	9
G	4	3	7
H	$\max(6,9)=9$	5	14
I	8	5	13
J	$\max(7,14)=14$	3	17

Project Duration: 17 days (EF of final activity J)

Step 2: Backward Pass (Calculate LF and LS)

Starting from the end (LF of J = 17):

Activity	LF	Duration	LS
J	17	3	14
I	17	5	12
H	14	5	9
G	14	3	11
F	9	5	4
E	9	3	6
D	12	2	10
C	min(4,11)=4	4	0
B	6	3	3
A	10	6	4

Step 3: Calculate Slack

Activity	ES	LS	EF	LF	Slack
A	0	4	6	10	4
B	0	3	3	6	3
C	0	0	4	4	0
D	6	10	8	12	4
E	3	6	6	9	3
F	4	4	9	9	0
G	4	11	7	14	7
H	9	9	14	14	0
I	8	12	13	17	4
J	14	14	17	17	0

Step 4: Identify Critical Path

Activities with Slack = 0 form the critical path:

Critical Path: C → F → H → J

Total Duration: 4 + 5 + 5 + 3 = 17 days

Path Analysis:

- Path 1 (A-D-I): $6 + 2 + 5 = 13$ days
- Path 2 (B-E-H-J): $3 + 3 + 5 + 3 = 14$ days
- Path 3 (C-F-H-J): $4 + 5 + 5 + 3 = 17$ days (Critical Path)
- Path 4 (C-G-J): $4 + 3 + 3 = 10$ days

Interpretation: Any delay in activities C, F, H, or J will delay the entire project. Activities A, B, D, E, G, and I have slack time and can be delayed without affecting the project completion date.

2.8 Gantt Chart

Definition: A Gantt Chart is a visual bar chart representing project schedule, showing tasks, durations, start/end dates, and dependencies.

2.8.1 Key Components

1. **Timeline:** Horizontal axis (days, weeks, months)
 2. **Tasks/Activities:** Vertical axis listing all project tasks
 3. **Bars:** Horizontal bars showing task duration and timing
 4. **Dependencies:** Arrows indicating task relationships
 5. **Milestones:** Special symbols marking key events
 6. **Progress Tracking:** Shaded portions showing completion percentage
-

2.9 Annual Profit and Related Calculations

Annual Profit Formula:

$$\text{Annual Profit} = \frac{\text{Total Income} - \text{Total Expenses}}{\text{Number of Years}}$$

Example 2.12: Annual Profit Calculation

Project Data:

- Total Income over 5 years: \$500,000
- Total Expenses over 5 years: \$350,000

Solution:

$$\text{Total Profit} = 500,000 - 350,000 = 150,000$$

$$\text{Annual Profit} = \frac{150,000}{5} = 30,000$$

Net Profit Calculation:

$$\text{Net Profit} = \text{Total Revenue} - \text{Total Costs}$$

2.10 Budgeting and Cost Management

2.10.1 Cost Categories

Direct Costs:

- Personnel costs (salaries, benefits)
- Materials and supplies
- Hardware and software
- Travel and meetings
- Consultants and subcontracting

Indirect Costs:

- General overheads (rent, utilities)
- Administrative staff
- Project overheads

2.10.2 Budget Formula

$$\text{Total Budget} = \text{Direct Costs} + \text{Indirect Costs} + \text{Contingency F}$$

Chapter 3: Risk and Quality Management

3.1 Risk Management

Definition: Risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on project objectives.

3.1.1 Risk Management Process

1. **Define Risk Management Standards**
2. **Identify Risks**
3. **Classify Risks**
4. **Develop Management Strategy**
5. **Monitor Risks**

3.1.2 Common Software Project Risks (Barry Boehm's List)

1. Personnel shortfalls
 2. Unrealistic schedules and budgets
 3. Incorrect requirements
 4. Gold-plating (unnecessary features)
 5. Scope creep
 6. Subcontractor issues
 7. Performance shortfalls
-

3.2 Risk Classification

3.2.1 Qualitative Risk Classification

Risks are classified based on:

- **Probability:** Likelihood of occurrence (scale: 1-5)
- **Impact:** Consequences if it occurs (scale: 1-5)

Probability Scale:

1. Very Low (0-10%)

2. Low (10-30%)
3. Medium (30-50%)
4. High (50-70%)
5. Very High (70-100%)

Impact Scale:

1. Very Low (minimal effect)
2. Low (minor effect)
3. Medium (moderate effect)
4. High (significant effect)
5. Very High (severe effect)

3.2.2 Risk Matrix

A two-dimensional matrix visualizing probability vs. impact:

Probability \ Impact	VL	L	M	H	VH
Very High (5)	Medium	High	High	Extreme	Extreme
High (4)	Low	Medium	High	High	Extreme
Medium (3)	Low	Medium	Medium	High	High
Low (2)	Low	Low	Medium	Medium	High
Very Low (1)	Low	Low	Low	Medium	Medium

3.2.3 Risk Weight Calculation

Formula:

$$\text{Risk Weight} = \text{Probability} \times \text{Impact}$$

Example 3.1: Risk Weight Calculation

Risk	Probability	Impact	Weight
Risk 1	2	2	4
Risk 2	5	3	15
Risk 3	3	5	15
Risk 4	4	4	16

Interpretation: Risk 4 has the highest weight (16) and should be prioritized for management.

3.3 Risk Zones

Risks are classified into three zones based on severity:

- **Green Zone:** Acceptable risks (Weight: 1-8)
 - **Yellow Zone:** Risks requiring monitoring and management (Weight: 9-15)
 - **Red Zone:** Unacceptable risks requiring avoidance (Weight: 16-25)
-

3.4 Expected Monetary Value (EMV)

Definition: EMV quantifies the financial impact of a risk by multiplying the probability of occurrence by the financial impact.

Formula:

$$\text{EMV} = \text{Probability} \times \text{Impact (in monetary terms)}$$

Example 3.2: EMV Calculation for Risk Budget

Risk	Probability	Impact (\$)	EMV (\$)
R1	0.30	1,000	300
R2	0.10	500	50
R3	0.50	200	100
R4	0.20	800	160
R5	0.40	300	120
Total EMV:			730

Solution:

$$\text{Total EMV} = 300 + 50 + 100 + 160 + 120 = 730$$

Interpretation: Allocate \$730 to the risk budget to manage these green zone risks. This amount covers R3 and R5 completely and most

of R1.

Example 3.3: Decision-Making with EMV

Scenario: Choose between two project approaches:

Approach A:

- Cost: \$50,000
- Success Probability: 70%
- Success Revenue: \$100,000
- Failure Revenue: \$0

Solution:

$$EMV_A = (0.70 \times 100,000) + (0.30 \times 0) - 50,000$$

$$EMV_A = 70,000 - 50,000 = 20,000$$

Approach B:

- Cost: \$30,000
- Success Probability: 50%
- Success Revenue: \$80,000
- Failure Revenue: \$0

Solution:

$$EMV_B = (0.50 \times 80,000) + (0.50 \times 0) - 30,000$$

$$EMV_B = 40,000 - 30,000 = 10,000$$

Interpretation: Approach A has higher EMV (\$20,000 vs. \$10,000) and is the better choice.

3.5 Risk Management Strategies

3.5.1 Negative Risk Strategies

1. **Avoidance:** Eliminate the risk by changing the project plan
2. **Mitigation:** Reduce probability or impact of the risk
3. **Transferral:** Shift risk to another party (e.g., insurance)
4. **Acceptance:** Acknowledge the risk and prepare contingency plans

3.5.2 Positive Risk (Opportunity) Strategies

1. **Exploitation:** Ensure the opportunity definitely happens
2. **Enhancement:** Increase probability or benefit of opportunity
3. **Sharing:** Partner with others to improve outcome
4. **Acceptance:** Recognize opportunity but don't change plan proactively

3.6 Quality Management

Definition: Quality management ensures software products meet necessary standards, requirements, and user expectations.

3.6.1 Quality Management Components

1. **Quality Planning:** Identifies relevant standards and implementation methods
2. **Quality Assurance (QA):** Ensures project follows quality standards
3. **Quality Control (QC):** Ensures products meet quality standards

3.6.2 Quality vs. Control

Quality Assurance (QA)	Quality Control (QC)
Process-focused	Product-focused
Preventive (defect prevention)	Detective (defect detection)
Proactive activities	Reactive activities
Code reviews, audits	Testing, inspections
Ensures correct process	Ensures correct product

3.7 Quality Metrics

3.7.1 Product Quality Metrics

- **Defect Density:** Number of defects per KLOC
- **Mean Time to Failure (MTTF):** Average operational time without failure
- **Reliability:** Probability of failure-free operation
- **Code Coverage:** Percentage of code tested

Defect Density Formula:

$$\text{Defect Density} = \frac{\text{Number of Defects}}{\text{KLOC}}$$

Example 3.4: Defect Density Calculation

- Total Defects: 45
- Lines of Code: 15,000 (15 KLOC)

Solution:

$$\text{Defect Density} = \frac{45}{15} = 3 \text{ defects per KLOC}$$

3.7.2 Process Quality Metrics

- **Defect Removal Efficiency (DRE):** Effectiveness of defect detection
- **Schedule Variance (SV):** Planned vs. actual schedule
- **Cost Variance (CV):** Planned vs. actual costs

DRE Formula:

$$\text{DRE} = \frac{\text{Defects Found Before Release}}{\text{Total Defects}} \times 100\%$$

Example 3.5: DRE Calculation

- Defects found during development: 80
- Defects found after release: 20
- Total defects: 100

Solution:

$$\text{DRE} = \frac{80}{100} \times 100\% = 80\%$$

Interpretation: 80% of defects were caught before release, indicating good quality control processes.

3.7.3 Code Quality Metrics

Cyclomatic Complexity: Measures program control flow complexity

$$\text{Cyclomatic Complexity} = E - N + 2P$$

Where:

- E = Number of edges in control flow graph
- N = Number of nodes
- P = Number of connected components (usually 1)

Simplified Formula:

$$\text{Cyclomatic Complexity} = \text{Number of Decision Points} + 1$$

Example 3.6: Cyclomatic Complexity

A function contains:

- 3 if statements
- 2 for loops
- 1 while loop

Solution:

$$\text{Cyclomatic Complexity} = 6 + 1 = 7$$

Interpretation:

- 1-10: Simple, low risk
 - 11-20: Moderate complexity
 - 21-50: Complex, high risk
 - >50 : Very high risk, difficult to test
-

Summary of Key Formulas

Financial Analysis

Metric	Formula
CBR	$\frac{\text{Total Benefits}}{\text{Total Costs}}$
BEP (Units)	$\frac{\text{Fixed Costs}}{\text{Price} - \text{Variable Cost}}$
NPV	$\sum_{t=0}^n \frac{B_t - C_t}{(1+r)^t}$
ROI	$\frac{\text{Annual Profit}}{\text{Total Expenses}} \times 100\%$
Annual Profit	$\frac{\text{Total Profit}}{\text{Years}}$

Estimation Methods

Method	Formula
PERT	$\frac{t_O + 4t_M + t_P}{6}$
COCOMO PM	$A_{PM} \times (\text{KSLOC})^{B_{PM}}$
COCOMO TDEV	$A_{TDEV} \times (\text{PM})^{B_{TDEV}}$
Function Points (UFP)	$\sum (\text{Count}_i \times \text{Weight}_i)$
VAF	$\frac{65 + \sum \text{Influence}_i}{100}$

Critical Path

Component	Formula
EF	ES + Duration
ES (multiple pred.)	max(EF of predecessors)
LS	LF – Duration
LF (multiple succ.)	min(LS of successors)
Slack	LS – ES or LF – EF

Risk Management

Metric	Formula
Risk Weight	Probability $\times$ Impact
EMV	Probability $\times$ Impact (\$)

Quality Metrics

Metric	Formula
Defect Density	$\frac{\text{Defects}}{\text{KLOC}}$
DRE	$\frac{\text{Defects Before Release}}{\text{Total Defects}} \times 100\%$
Cyclomatic Complexity	Decision Points + 1

Exam Preparation Tips

1. **Memorize all formulas** listed in the summary section
 2. **Practice calculations** for NPV, COCOMO, and Critical Path
 3. **Understand concepts** behind each metric, not just formulas
 4. **Work through examples** multiple times with different values
 5. **Create formula sheet** for quick reference during exam preparation
 6. **Focus on interpretation** of calculated results
 7. **Review risk classification** matrices and zones
 8. **Understand differences** between QA and QC
 9. **Practice SWOT analysis** and Score Matrix evaluations
 10. **Master critical path** forward and backward calculations
-

Practice Problems

Problem 1: Comprehensive Feasibility Analysis

A software company proposes developing a mobile application:

- Development Cost: \$120,000
- Expected Revenue Year 1: \$30,000
- Expected Revenue Year 2: \$50,000

- Expected Revenue Year 3: \$60,000
- Discount Rate: 8%
- Fixed Costs: \$40,000
- Variable Cost per User: \$2
- Selling Price per User: \$10

Calculate:

1. NPV
2. CBR
3. BEP (users)
4. ROI

Problem 2: COCOMO Estimation

Estimate effort, time, and team size for:

- Project Size: 15 KSLOC
- Project Type: Semi-Detached

Problem 3: Critical Path Analysis

Given activities A-J with durations and dependencies (as in Example 2.11), identify:

1. Critical path
2. Project duration
3. Slack for each activity

Problem 4: Risk EMV Calculation

Calculate total EMV for 5 risks with given probabilities and impacts, and recommend risk budget allocation.

Conclusion

This comprehensive study guide covers all essential topics for your MSc Software Project Management course:

- **Chapter 1:** Feasibility study types and economic analysis methods (CBR, NPV, BEP)

- **Chapter 2:** Project management techniques (Function Points, COCOMO, ROI, Critical Path, Gantt Charts)
- **Chapter 3:** Risk classification, EMV calculations, risk zones, and quality management metrics

All formulas include complete mathematical derivations and practical examples with step-by-step solutions. Use this guide for comprehensive exam preparation and practical application in software project management.

Good luck with your 3-credit MSc examination!